

Concentration and profile of 22 urinary polycyclic aromatic hydrocarbon metabolites in the US population.

Urinary monohydroxy polycyclic aromatic hydrocarbons (OH-PAHs) are a class of PAH metabolites used as biomarkers for assessing human exposure to PAHs. The Centers for Disease Control and Prevention's National Health and Nutrition Examination Survey (NHANES) uses OH-PAHs to establish reference range concentrations for the US population, and to set benchmarks for future epidemiologic and biomonitoring studies. For the years 2001 and 2002, 22 OH-PAH metabolites were measured in urine specimens from 2748 NHANES participants. Percentages of samples with detectable levels ranged from nearly 100% for metabolites of naphthalene, fluorene, phenanthrene, and pyrene, to less than 5% for metabolites from parent compounds with higher molecular weight such as chrysene, benzo[c]phenanthrene, and benz[a]anthracene. The geometric mean for 1-hydroxypyrene (1-PYR)--the most commonly used biomarker for PAH exposure--was 49.6 ng/L urine, or 46.4 ng/g creatinine. Children (ages 6-11) generally had higher levels than did adolescents (ages 12-19) or adults (ages 20 and older). Model-adjusted, least-square geometric means for 1-PYR were 87, 53 and 43 ng/L for children, adolescents (ages 12-19) and adults (ages 20 years and older), respectively. Log-transformed concentrations for major detectable OH-PAHs were significantly correlated with each other. The correlation coefficients between 1-PYR and other metabolites ranging from 0.17 to 0.63 support the use of 1-PYR as a useful surrogate representing PAH exposure.